

UNIT 2 Welcome to a Modern Port

Today a major port is a world of its own. Modern port facilities have to meet the demands of highly specialised transport systems by sea, land and air. And so within the boundaries of any large sea port we find an amazing range of services.

A modern port maintains shipping channels, harbour basins, and navigation aids as well as the port infrastructure such as wharves, quays, docks, storage areas and warehouses, cranes for cargo handling and terminals for cargoes and passengers. It provides pilots and pilot vessels, tugs for towage and emergency response vessels in cases of accidents at sea.

Within its area we find office buildings for all the various companies catering for the shipping industry - Ship Brokers, Shipping Agents, Stevedoring and Lashing Companies, Ship's Chandlers, Marine Engineering Companies, Companies for Marine Equipment and Ship's Repair, Cargo Surveyors and Inspection Companies, to name only a few. Today there are also strict security requirements.

The Port of Santander

Elegant, picturesque Santander, situated in the region of Cantabria, is typical of North Spain. The city lies in a beautiful setting, backed by open countryside and dramatic mountains. A very good road network connects Santander to the whole of Spain, Portugal and the South of France. A bustling regional capital and a popular holiday resort, Santander offers excellent beaches, splendid viewpoints, museums, galleries and a lively market.

The port activity in Santander dates back more than 2000 years, the town being founded by the Romans in 19 BC.

Today the Port of Santander is a fast-growing port and an important junction in Spain for combined water, road and railway transport. In 2003 the total port traffic comprised about five million tons of goods. The Port of Santander has facilities for handling all types of cargo: dry and liquid bulk, general cargo, RORO and container traffic. The passenger terminal in the heart of the city has facilities for both ferries and cruise ships and there is regular ferry liner traffic to the United Kingdom.

Four million tonnes of dry bulk such as coal, ore, fertilisers, grain, soya beans, cereals and sodium carbonate passed through Santander in 2003. More than 1 million tons of general cargo was handled, RORO traffic accounted for 65% of this type of cargo. Of liquid bulk petroleum is the largest element, followed by chemical products and molasses.

Depending on the type, bulk goods can be loaded and unloaded using a conveyor, grabber or bucket. The bulk is loaded into a hopper, which passes it onto a conveyor belt system to deliver it to a stockpile.

Some bulk cargo such as scrap metal is loaded into big metal bins known as skips and lifted into the hold of the bulk carrier.

During the last few years vehicle handling shows the greatest increase in Santander: 36% in 2003. This is important for the port's image, as shipment of motor vehicles requires excellent standards of logistics operators, as well as port facilities and shipping lines.

The port has invested in and is continually making new investments in specialised handling installations for all the types of cargo already mentioned.

Reading Comprehension: Vessels and Cargoes

The general cargo ship

During the last few decades shipping has seen a great deal of change. Until the 20th century the most important cargo ship was the break-bulk carrier, sometimes called the general cargo ship or freighter. The cargo holds on these ships could carry almost any kind of cargo, both piece goods and bulk cargo. The cargo was packed into drums, boxes, bags, bales and crates or on pallets. The ship was loaded and unloaded using portside cranes and ship's derricks that lifted the cargo through the hatches and stored it into the holds. Cargo could also be secured by lashing and stored on deck.

The bulk carrier

Today modern commercial vessels are highly specialised, designed to carry specific types of cargo. The names of the ships tell us what type of cargo they are designed to carry. The bulk carrier carries bulk cargo, "loose" cargo, either "dry bulk" such as coal, grain, iron ore, fertilisers or "liquid bulk" such as a range of chemicals including petroleum products. Bulk carriers have huge under-deck specialised holds where the bulk products are poured and stored. Bulk carriers come in different sizes, from the so-called "handysize" bulk carriers of about 25 000 DWT to very large carriers of up to 200 000 DWT.

The multi-purpose carrier

The MS Marina, the ship in our story, is a type of multi-purpose dry cargo carrier. "Multi-Purpose Carriers" transport different kinds of cargo: break-bulk and/or pure bulk cargoes and/or containers, even reefer containers for frozen meats or fruit. Many of these carriers are so-called geared vessels, they are equipped with their own cargo cranes for loading and discharging. Hatch openings are designed to fit standard container sizes. Removable between decks or ('tweendecks) increase the number of available holds. In the huge bulk holds there are removable bulkheads that help prevent the grain and other bulk cargoes from shifting.

The container ship

It was in the 1960s that the first container ship was built and since then she has revolutionised shipping. A container ship is designed to carry cargo in thousands of standard-size boxes - "containers", either 20 ft units or 40 ft units. At the container

terminals these ships can quickly load and discharge by means of large quay-side cranes called portainers or gantry cranes. The cranes lift the containers off or onto the quay or trucks and off or onto the ship's deck. While a conventional dry cargo vessel may take 3-4 days to load or discharge, a container ship can do the same in a matter of hours. Today container ships are seen in ports all over the world and are gradually replacing the general cargo ships.

Seen from a distance the container ship has a very characteristic "flat" silhouette. The small superstructure with the navigating bridge is at the stern of the ship and the many containers are stacked fore of the bridge along the whole length of the vessel.

The reefer

The Refrigerated Cargo Carrying Vessels, reefers, are built to carry fruit, meat, vegetables and other refrigerated food products that require refrigerating equipment to stay fresh during a sea voyage. The cargo can be stowed on pallets in the refrigerated holds. Some reefers carry reefer containers.

Perhaps the most famous of these types of vessels are the banana carriers, trading between the Caribbean and Europe.

The RORO vessel

Another special cargo vessel is known as the RORO, which means a roll-on, roll-off vessel. The RORO ships have huge stern (or bow) doors which are lowered to make a bridge from the ship to the wharf. On the RORO-vessel whole trucks can roll on and off very rapidly at ports. RORO is often used to carry large numbers of cars/motor vehicles, which are rapidly loaded via a stern ramp.

The wharf machinery can also get into the ship's interior and collect cargo to bring it ashore.

The roll-on, roll-off vessel comes in two main types: the passenger RORO and the cargo RORO. Modern car-passenger ferries take care of most passenger traffic on short-sea routes in the Baltic Sea and the North Sea. It is probably the only type of cargo vessel that most people have travelled on. The vehicles are driven on and off through a stern door (sometimes a bow door) and stored on the car deck below the passenger accommodation areas. Since the Estonia disaster, when the bow visor gave way in the heavy storm and the car deck was flooded, safety regulations on

board car- passenger ferries have become stricter and the door structures have improved.

The oil tanker

One of the most important vessels in the world's merchant fleets today is the tanker. The growth in size of the tankers has been extremely rapid during recent years. Tankers carry liquid cargo, not only oil, in tanks. The best known are the oil tankers. They come in two kinds: the crude carrier, which carries crude oil, and the clean products tanker, which carries refined products such as petrol, gasoline, aviation fuel, kerosene and paraffin. Tankers range in all sizes from small bunkering tankers of 1000 DWT, used for refuelling larger vessels, to the real giants the VLCC (Very Large Crude Carrier) of 200 000 - 300 000 DWT and the ULCC (Ultra Large Crude Carrier) of over 300 000 DWT.

In 1990, the United States enacted the Oil Pollution Act (OPA). It requires a gradual introduction of tankers with double hulls, i.e. not just with double bottoms but also double on both sides. Full compliance with the law is as far away as 2015. This stamp shows a picture of the modern double-hulled Mobil Oil tanker 'Eagle', built 1993, of 284,493 DWT. However, most tankers trading worldwide today are still single-hulled vessels.

The LNG carrier

(Liquified Natural Gas) and the LPG carrier (Liquified Petroleum Gas) are really a kind of tanker and came in in the nineteen-hundreds. LNG and LPG are not the easiest of cargoes to transport. In its natural state, LNG is a gas; it must be pressurised into being liquid or kept as a liquid by reducing the temperature in order to transport it. The shape of the LNG Carrier, with the huge round Moss tanks seen along the deck, has led to the nickname of "Dinosaur Eggs Carriers".

The LNG vessels carry explosive gas - kept at below freezing temperatures - as an unstable liquid. This means that they carry extremely dangerous cargo. The vessels must be maintained very thoroughly and safety measures on board have to be very strict.

The Lash

Another kind of cargo carrier is the Lash. LASH means lighter aboard ship (lighter = barge). The vessel has a huge 500 ton crane on the main deck. The holds are divided into cells to make room for the LASH barges which the crane plucks from the water at the stern of the ship, carries along the deck and stows in the ship's cells for the voyage. LASH barges are loaded at inland river and ocean ports. The barges are then towed to meet the LASH mother vessel and lifted aboard. When the mother vessel arrives at its port of destination the huge crane lowers the LASH barges into the water, where they are then towed to their final destination. A Lash ship does not need to tie up to a port during discharging.

Heavy-lift ships

Another very special type of ship is the heavy-lift ship. This type is a little like the LASH. The heavy-lift ship is designed to carry exceptionally heavy loads and unusual cargoes such as power plants, oil rigs, generators, and yachts. The cargo (even small vessels) can be stowed aboard the ships by a lift-on lift-off method using the ship's own cranes or by a float-in float-out method where the mother ship is partly submerged under water during loading and unloading the cargo she is to carry.